OS Skill-22

To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings.

To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate Reports.

```
lahari@DESKTOP-AIC33F7:~$ sudo apt update
sudo apt install gcc valgrind -y
[sudo] password for lahari:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1122 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1397 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [228 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1220 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [245 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.2 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1674 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [308 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1702 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [383 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [45.4 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [12.1 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [342 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1805 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [411 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [50.7 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [14.0 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5784 B]
Get:27 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [748 B]
Fetched 12.0 MB in 4s (2719 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
124 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gcc is already the newest version (4:13.2.0-7ubuntu1).
valgrind is already the newest version (1:3.22.0-0ubuntu3).
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 124 not upgraded.
```

nano memory.c

```
#include <stdio.h>
#include <stdlib.h>

int main() {
   int *ptr = malloc(5 * sizeof(int));
```

```
    if (ptr == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < 5; i++) {
        ptr[i] = i + 1;
    }

    printf("Values: ");
    for (int i = 0; i < 5; i++) {
        printf("%d ", ptr[i]);
    }
    printf("\n");

    // Intentionally not freeing memory
    return 0;
}
nano calculator.c

#include <stdio.h>

int main() {
    int a, b;

    if (scanf("%d %d", &a, &b) != 2) {
        fprintf(stderr, "Invalid input\n");
        return 1;
    }

    printf("%d\n", a + b);

    return 0;
}
```

```
lahari@DESKTOP-AIC33F7:~$ gcc --version
valgrind --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
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This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

valgrind-3.22.0
lahari@DESKTOP-AIC33F7:~$ mkdir valgrind_lab
cd valgrind_lab
nano memory.c
lahari@DESKTOP-AIC33F7:~/valgrind_lab$ gcc -g -Wall -Wextra memory.c -o memory
lahari@DESKTOP-AIC33F7:~/valgrind_lab$ valgrind --leak-check=full --show-leak-kinds=all ./memory
==736== Memcheck, a memory error detector
==736== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==736== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==736== Command: ./memory
==736==
Values: 1 2 3 4 5
==736==
==736== HEAP SUMMARY:
==736==     in use at exit: 20 bytes in 1 blocks
==736==   total heap usage: 2 allocs, 1 frees, 1,044 bytes allocated
==736==
==736== 20 bytes in 1 blocks are definitely lost in loss record 1 of 1
==736==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==736==    by 0x1091BE: main (memory.c:6)
==736==
==736== LEAK SUMMARY:
==736==    definitely lost: 20 bytes in 1 blocks
==736==    indirectly lost: 0 bytes in 0 blocks
==736==      possibly lost: 0 bytes in 0 blocks
==736==    still reachable: 0 bytes in 0 blocks
==736==         suppressed: 0 bytes in 0 blocks
==736==
==736== For lists of detected and suppressed errors, rerun with: -s
==736== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
lahari@DESKTOP-AIC33F7:~/valgrind_lab$ nano memory.c
lahari@DESKTOP-AIC33F7:~/valgrind_lab$ gcc -g -Wall -Wextra memory.c -o memory
valgrind --leak-check=full --show-leak-kinds=all ./memory
```

```
lahari@DESKTOP-AIC33F7:~/valgrind_lab$ nano memory.c
lahari@DESKTOP-AIC33F7:~/valgrind_lab$ gcc -g -Wall -Wextra memory.c -o memory
valgrind --leak-check=full --show-leak-kinds=all ./memory
==745== Memcheck, a memory error detector
==745== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==745== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==745== Command: ./memory
==745==
Values: 1 2 3 4 5
==745==
==745== HEAP SUMMARY:
==745==     in use at exit: 0 bytes in 0 blocks
==745==   total heap usage: 2 allocs, 2 frees, 1,044 bytes allocated
==745==
==745== All heap blocks were freed -- no leaks are possible
==745==
==745== For lists of detected and suppressed errors, rerun with: -s
==745== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
lahari@DESKTOP-AIC33F7:~/valgrind_lab$ valgrind --leak-check=full --show-leak-kinds=all ./memory 2> valgrind_report.txt
Values: 1 2 3 4 5
lahari@DESKTOP-AIC33F7:~/valgrind_lab$ cat valgrind_report.txt
==746== Memcheck, a memory error detector
==746== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==746== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==746== Command: ./memory
==746==
==746==
==746== HEAP SUMMARY:
==746==     in use at exit: 0 bytes in 0 blocks
==746==   total heap usage: 2 allocs, 2 frees, 1,044 bytes allocated
==746==
==746== All heap blocks were freed -- no leaks are possible
==746==
==746== For lists of detected and suppressed errors, rerun with: -s
==746== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

nano test.sh

```
#!/bin/bash

gcc -Wall -Wextra calculator.c -o calculator || exit 1

passed=0
failed=0

echo "Automated Test Report"
echo "---------------------"

run_test() {
  name="$1"
  input="$2"
  expected="$3"

  actual=$(printf '%s\n' "$input" | ./calculator 2>/dev/null)
  status=$?

  if [ "$status" -eq 0 ] && [ "$actual" = "$expected" ]; then
    echo "PASS: $name"
    passed=$((passed + 1))
  else
```

```
        echo "FAIL: $name"
        echo "  Expected: $expected"
        echo "  Actual:   $actual"
        failed=$((failed + 1))
    fi
}

run_test "Positive numbers" "10 20" "30"
run_test "Negative numbers" "-5 -3" "-8"
run_test "Zero values" "0 0" "0"
run_test "Mixed numbers" "-5 10" "5"

echo "----------------------"
echo "Passed: $passed"
echo "Failed: $failed"
echo "Total:  $((passed + failed))"

if [ "$failed" -eq 0 ]; then
    echo "Result: ALL TESTS PASSED"
else
    echo "Result: SOME TESTS FAILED"
fi

exit "$failed"
```

```
==748== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
lahari@DESKTOP-AIC33F7:~/valgrind_lab$ mkdir testing_lab
cd testing_lab
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ nano calculator.c
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ gcc -Wall -Wextra calculator.c -o calculator
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ ls
calculator  calculator.c
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ nano test.sh
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ chmod +x test.sh
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ ./test.sh
Automated Test Report
--------------------
PASS: Positive numbers
PASS: Negative numbers
PASS: Zero values
PASS: Mixed numbers
--------------------
Passed: 4
Failed: 0
Total:  4
Result: ALL TESTS PASSED
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ ./test.sh | tee test_report.txt
Automated Test Report
--------------------
PASS: Positive numbers
PASS: Negative numbers
PASS: Zero values
PASS: Mixed numbers
--------------------
Passed: 4
Failed: 0
Total:  4
Result: ALL TESTS PASSED
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ cat test_report.txt
Automated Test Report
--------------------
PASS: Positive numbers
PASS: Negative numbers
PASS: Zero values
PASS: Mixed numbers
--------------------
Passed: 4
Failed: 0
Total:  4
Result: ALL TESTS PASSED
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ nano test.sh
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ ./test.sh
Automated Test Report
----------------------
FAIL: Positive numbers
  Expected: 35
  Actual:   30
PASS: Negative numbers
PASS: Zero values
PASS: Mixed numbers
----------------------
Passed: 3
Failed: 1
Total:  4
Result: SOME TESTS FAILED
lahari@DESKTOP-AIC33F7:~/valgrind_lab/testing_lab$ ~
```